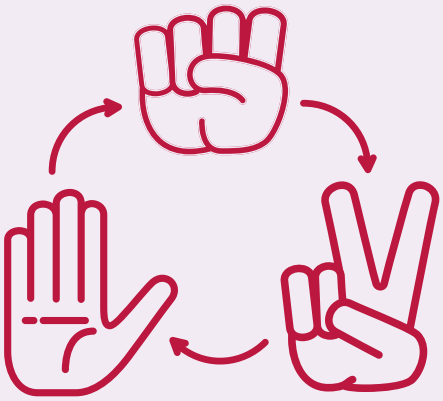

Rock, Paper, Scissors

Raeygan Bollaert

Final Logic Case Study

194506 Introduction to Front End Technologies

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Pseudo code

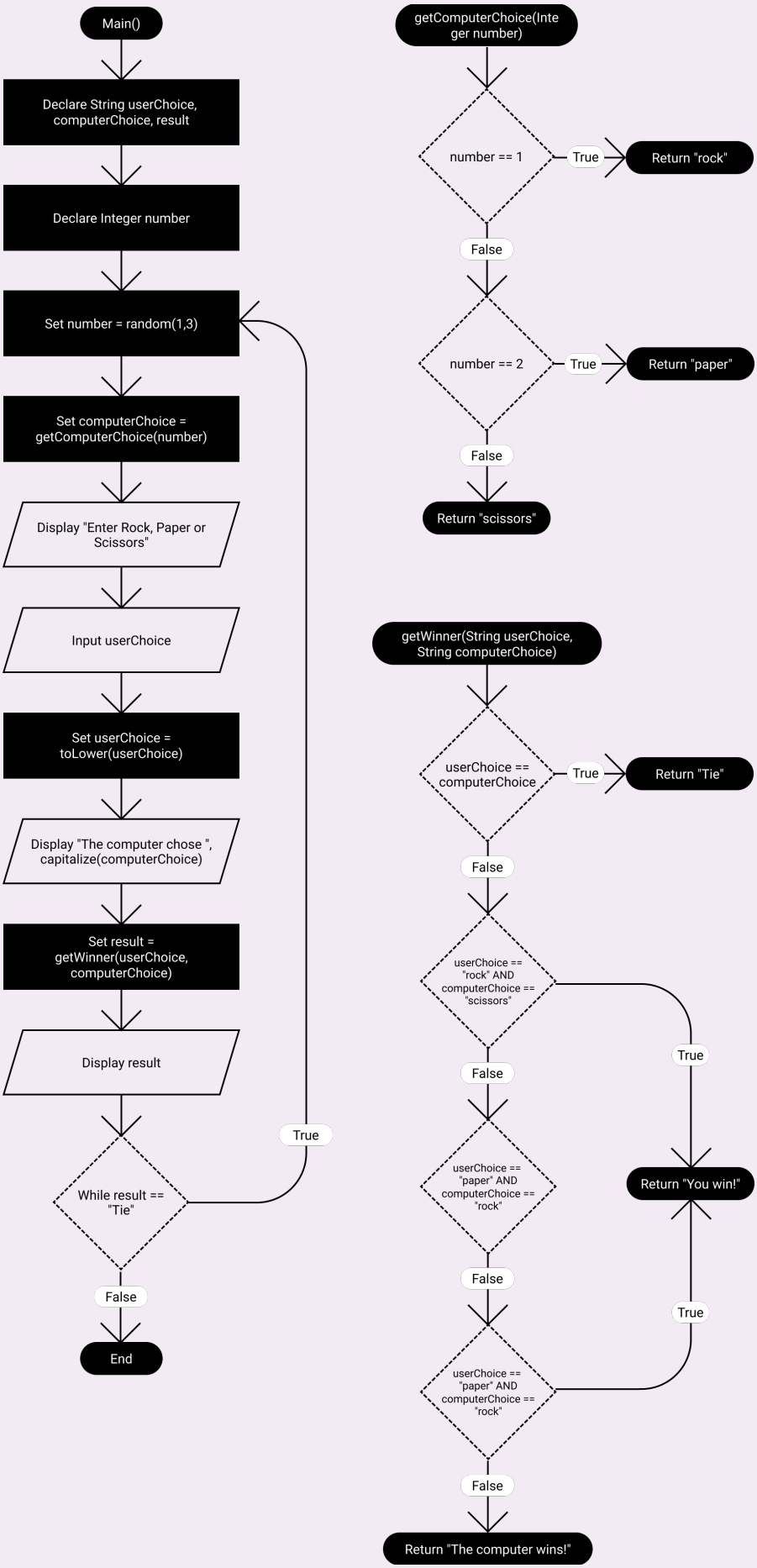
```
Module main()

  // Declare the variables
  Declare String userChoice, computerChoice, result
  Declare Integer number

  Do
    // Generate a random number
    Set number = random(1, 3)

    // Pass the number into function
    Set computerChoice = getComputerChoice(number)

    Display "Enter Rock, Paper or Scissors"
    Input userChoice

    // Standardise for comparison
    Set userChoice = toLower(userChoice)

    Display "The computer chose ", capitalize(computerChoice)

    // Decide the winner
    Set result = getWinner(userChoice, computerChoice)
    Display result

  // Repeat if the round is a tie
  While result == "Tie"

End Module


// Return the computer's choice based on the random number
Function String getComputerChoice(Integer number)

  If number == 1 Then
    Return "rock"

  Else If number == 2 Then
    Return "paper"

  Else
    Return "scissors"
```

// Continued on next page

Pseudo code (Continued)

```
// Continued from previous page

  End If

End Function


// Decide the winner
Function String getWinner(String userChoice, String computerChoice)

  If userChoice == computerChoice Then
    Return "Tie"

  Else If userChoice == "rock" AND computerChoice == "scissors" Then
    Return "You win!"

  Else If userChoice == "scissors" AND computerChoice == "paper" Then
    Return "You win!"

  Else If userChoice == "paper" AND computerChoice == "rock" Then
    Return "You win!"

  Else
    Return "The computer wins!"

  End If

End Function
```

Flowchart


Main()
Declare String userChoice, computerChoice, result
Declare Integer number
Set number = random(1,3)
Set computerChoice = getComputerChoice(number)
Display "Enter Rock, Paper or Scissors"
Input userChoice
Set userChoice = toLower(userChoice)
Display "The computer chose ", capitalize(computerChoice)
Set result = getWinner(userChoice, computerChoice)
Display result
While result == "Tie"
True
False
End
getComputerChoice(Integer number)
number == 1
True
Return "rock"
False
number == 2
True
Return "paper"
False
Return "scissors"
getWinner(String userChoice, String computerChoice)
userChoice == computerChoice
True
Return "Tie"
False
userChoice == "rock" AND computerChoice == "scissors"
True
False
userChoice == "paper" AND computerChoice == "rock"
Return "You win!"
False
True
userChoice == "paper" AND computerChoice == "rock"
False
Return "The computer wins!"

Python

```
import random

def main():
    play_again = "y"
    while play_again == "y":
        result = "Tie"

        while result == "Tie":
            number = random.randint(1, 3)
            computer_choice = get_computer_choice(number)
            user_choice = input("Enter Rock, Paper or Scissors: ").lower()
            print("The computer chose", computer_choice.capitalize())
            result = get_winner(user_choice, computer_choice)
            print(result)

        play_again = input("Do you want to play again? (y/n): ").lower()

def get_computer_choice(number):
    if number == 1:
        return "rock"
    elif number == 2:
        return "paper"
    else:
        return "scissors"

def get_winner(user_choice, computer_choice):
    if user_choice == computer_choice:
        return "Tie"
    elif user_choice == "rock" and computer_choice == "scissors":
        return "You win!"
    elif user_choice == "scissors" and computer_choice == "paper":
        return "You win!"
    elif user_choice == "paper" and computer_choice == "rock":
        return "You win!"
    else:
        return "The computer wins!"

main()
```

References

Textbook:
Gaddis, T. (2020). Starting out with programming logic and design (5th ed.). Pearson.

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https://thenounproject.com/icon/rock-papper-and-scissors-2837622/